

6.2 Suggested Project 2: Source Geolocation using Frequency Measurements

Project Summary

- Formulate the source geolocation problem under Gaussian measurement noise as a nonlinear least squares (NLS) problem.
- Derive the analytical expression for the gradient of the cost function and validate your results using numerical methods for computing the derivatives.
- Design and realize your own gradient-based algorithm with line search that takes into account the bound constraints on the design variables.
- Evaluate the geolocation accuracy of your method under different noise levels and compare it with at least one benchmark method at your choice.
- Analyze the efficiency of your method in terms of the number of iterations required by your gradient-based algorithm and the number of objective function evaluations required by your line search method.

6.2.1 Problem Description

Geolocation of radio sources on the Earth surface is a classic estimation problem, which has found numerous civil and military applications in navigation, search and rescue, and surveillance.

In this project, we shall consider the use of four satellites to locate a ground radio source using the frequency difference of arrival (FDOA) measurements only. This is known in some literature as FDOA-only localization, which is a recently exploited topic. For example, see the paper below and references there in:

[1]. Y. Pei, X. Li, Le Yang and F. Guo, “A closed-form solution for source localization using FDOA measurements only,” *IEEE Communications Letters*, vol. 27, pp. 115-119, Jan. 2022.

6.2.2 Mathematical Background

In FDOA-only geolocation, each satellite in the satellite group receives the signal of the *same* radio source. The source is located at $\mathbf{u}^o = [x^o, y^o, z^o]^T$ in the Earth Centered Earth Fixed (ECEF) coordinate system. Its geodetic position is denoted as $\mathbf{p}^o = [B^o, L^o, H^o]^T$, where B^o , L^o and H^o represents the latitude, longitude and altitude. Since the radio source is assumed to be on the ground, we have $H^o = 0$.

The Cartesian coordinates $\mathbf{u}^o$ are related to $\mathbf{p}^o$ via

$$x^o = (R_E(B^o) + H)\cos(B^o)\cos(L^o), \quad (6.1a)$$

$$y^o = (R_E(B^o) + H)\cos(B^o)\sin(L^o), \quad (6.1b)$$

$$z^o = [(1 - e^2)R_E(B^o) + H]\sin(B^o), \quad (6.1c)$$

where

$$R_E(B^o) = \frac{R_0}{\sqrt{1 - e^2 \sin^2(B^o)}}, \quad (6.2)$$

$e = 0.081819198425$ is the eccentricity, and $R_0 = 6378.137\text{km}$ is the Earth equatorial radius. For MATLAB functions on coordinate conversion, see

- <https://www.mathworks.com/help/map/ref/geodetic2ecef.html>,
- <https://www.mathworks.com/help/map/ref/ecef2geodetic.html>.

Let us denote the position and velocity of satellite i , $i = 1, 2, 3, 4$, in the ECEF coordinate system as $\mathbf{s}_i = [s_{x,i}, s_{y,i}, s_{z,i}]^T$ and $\dot{\mathbf{s}}_i = [\dot{s}_{x,i}, \dot{s}_{y,i}, \dot{s}_{z,i}]^T$. Suppose the radio source is transmitting with an unknown frequency f_0 . The signal frequency measured at satellite i is thus subject to the Doppler effect and is equal to

$$f_i = f_0 + \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_i)^T \dot{\mathbf{s}}_i}{\|\mathbf{u}^o - \mathbf{s}_i\|} + n_i, \quad (6.3)$$

where $\lambda = \frac{c}{f_0}$ is the source signal wavelength, and n_i is the Gaussian random noise with zero mean and variance σ_f (i.e., $n_i \sim \mathcal{N}(0, \sigma_f^2)$).

Since the source signal frequency is not known, we subtract the received frequency at satellite 1 (also called the reference satellite) from the received frequencies at other satellites to obtain the FDOA measurements¹. Specifically, the measured FDOA f_{j1} between the satellite pair j and 1, $j = 2, 3, 4$, is

$$f_{j1} = f_j - f_1 = \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_j)^T \dot{\mathbf{s}}_j}{\|\mathbf{u}^o - \mathbf{s}_j\|} - \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_1)^T \dot{\mathbf{s}}_1}{\|\mathbf{u}^o - \mathbf{s}_1\|} + (n_j - n_1). \quad (6.4)$$

Collecting f_{j1} for $j = 2, 3, 4$ generates the measurement vector $\mathbf{f} = [f_{21}, f_{31}, f_{41}]^T$. Besides, define $\mathbf{g}(B^o, L^o)$ as the true FDOA function

$$\mathbf{g}(B^o, L^o) = \begin{bmatrix} g(B^o, L^o, \mathbf{s}_2, \dot{\mathbf{s}}_2) - g(B^o, L^o, \mathbf{s}_1, \dot{\mathbf{s}}_1) \\ g(B^o, L^o, \mathbf{s}_3, \dot{\mathbf{s}}_3) - g(B^o, L^o, \mathbf{s}_1, \dot{\mathbf{s}}_1) \\ g(B^o, L^o, \mathbf{s}_4, \dot{\mathbf{s}}_4) - g(B^o, L^o, \mathbf{s}_1, \dot{\mathbf{s}}_1) \end{bmatrix}, \quad (6.5)$$

where from (6.4),

$$g(B^o, L^o, \mathbf{s}_i, \dot{\mathbf{s}}_i) = \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_i)^T \dot{\mathbf{s}}_i}{\|\mathbf{u}^o - \mathbf{s}_i\|}. \quad (6.6)$$

The measurement noise in $\mathbf{f}$ is thus equal to $\mathbf{n}_f$, which is

$$\mathbf{n}_f = \begin{bmatrix} n_2 - n_1 \\ n_3 - n_1 \\ n_4 - n_1 \end{bmatrix}. \quad (6.7)$$

We are interested in identifying the source position $\mathbf{p}^o$ (with its altitude $H^o = 0$) using the measured FDOAs in $\mathbf{f}$ as well as the known satellite positions $\mathbf{s}_i$ and velocities $\dot{\mathbf{s}}_i$, $i = 1, 2, 3, 4$.

¹This is how the name ‘FDOA’ comes from. We are using indeed the *difference* of the Doppler frequencies received at various satellites for source geolocation.

6.2.3 Guidelines for Progress Report

1). (30%) Show that the FDOA measurement vector $\mathbf{f}$ follows a multivariate Gaussian distribution with mean $\mathbf{g}(B^\circ, L^\circ)$ and covariance $\sigma_f^2 \mathbf{Q}$, where $\mathbf{Q}$ takes the form

$$\mathbf{Q} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}. \quad (6.8)$$

Then, formulate the FDOA-only geolocation problem using the maximum likelihood (ML) estimation principle as a nonlinear least-squares problem.

Hint: Refer to [Chapter 7](#) in this notes for an introduction to the multivariate Gaussian density and ML estimation.

2). (40%) Consider the following scenario. The radio source has a carrier frequency of $f_0 = 1\text{GHz}$. It is located at $\mathbf{p}^o = [5^\circ, 10^\circ, 0]^T$, where 5° means the north latitude of 5 degrees and 10° means the east longitude of 10 degrees. The satellite positions and velocities are listed below:

$$\begin{aligned} \mathbf{s}_1 &= [7378.1, \quad 0, \quad 0]^T \text{ km}, \dot{\mathbf{s}}_1 = [0.0001, 4.4995, 5.3623]^T \text{ km/s}, \\ \mathbf{s}_2 &= [7377.5, \quad 100, \quad 0]^T \text{ km}, \dot{\mathbf{s}}_2 = [-0.0671, 4.9493, 4.9497]^T \text{ km/s}, \\ \mathbf{s}_3 &= [7377.5, -100, \quad 0]^T \text{ km}, \dot{\mathbf{s}}_3 = [0.0610, 4.4991, 5.3623]^T \text{ km/s}, \\ \mathbf{s}_4 &= [7377.5, \quad 0, \quad 100]^T \text{ km}, \dot{\mathbf{s}}_4 = [-0.0777, 4.0150, 5.7335]^T \text{ km/s}. \end{aligned}$$

It is priorly known that the latitude and longitude of the radio source are within $[0, 40^\circ]$ and $[0, 40^\circ]$. Implement the grid search method for accomplishing geolocation. The grid search process is as follows. First, the priorly known location area of the radio source is covered with a uniform grid. We then evaluate the objective function at each grid point. The grid point with the minimum objective function value is output as the geolocation result.

In the progress report, assume that the signal frequency of the radio source is *known*.

3). (30%) Implement a Monte Carlo simulation program to investigate the geolocation accuracy of your grid search method in terms of estimation root mean square error (RMSE) under different FDOA noise levels ($\sigma_f = 1, 2, 4, 8, 16\text{Hz}$).

Suppose in the l -th Monte Carlo run, the estimated source position in the ECEF coordinates is denoted by $\mathbf{u}_l$. The geolocation RMSE is computed using

$$\text{RMSE} = \sqrt{\frac{1}{L} \sum_{l=1}^L \|\mathbf{u}_l - \mathbf{u}^o\|^2}, \quad (6.9)$$

where L is the total number of Monte Carlo runs (normally set to be e.g, 10^4).

6.2.4 Guidelines for Final Report

1). Re-use your progress report that details the problem formulation and grid-based search method.

2). (10%) In order to generate the true FDOAs using (6.5), the signal wavelength needs to be provided. But it is still not known, as $\lambda = c/f_0$ requires the knowledge on the

unknown signal frequency f_0 . Find a way to obtain an estimate of the signal wavelength λ . Justify your design by carrying out an analysis on the approximation error.

Use this frequency estimate throughout the remaining experiments.

Bonus: In the current setup, we consider exploiting the FDOAs f_{21}, f_{31} and f_{41} only. The other possible FDOAs such as f_{23} and f_{34} are not used. Address the following question in your final report: Does utilizing these FDOAs bring any performance improvement? Justify your answer using theoretical analysis and you will receive 5% of your final report mark as bonus.

3). (30%) Derive with the help of chain rule the analytical expression of the gradient of the true FDOA $g(B^o, L^o, \mathbf{s}_j, \dot{\mathbf{s}}_j) - g(B^o, L^o, \mathbf{s}_1, \dot{\mathbf{s}}_1)$. Verify your result using the finite differences method for computing the gradient/derivatives (see [Chapter 4.4](#) of this note).

Bonus: if you can also numerically verify your results using the complex step method (see also [Chapter 4.4](#) of this note), you will receive 10% of your final report mark as bonus.

4). (35%) Select, modify and implement one of the following gradient-base methods **on your own**:

- Conjugate gradient
- Newton's method
- Gauss Newton method
- Levenberg-Marquardt algorithm
- BFGS algorithm

with line search and stopping criteria. See [Chapter 4.3](#) for an introduction to these gradient-based algorithms, line search and stopping criteria.

Make sure that in each iteration, the geolocation solution ALWAYS stays in the area of interest (i.e., the area with the latitude range $[0, 40^\circ]$ and longitude range $[0, 40^\circ]$).

In your implementation, also include a routine to realize the multistart strategy (i.e., generate multiple initial solutions such that multiple solution sequence can be produced using your gradient-based method. You can output the solution with the smallest objective function value as the final geolocation result). See [Chapter 4.3.6](#) for details.

5). (25%) Compare, using Monte Carlo simulation, the geolocation accuracy, in terms of estimation RMSE (see (6.9)), of your gradient-based algorithm against that of your grid search algorithm (now with the estimated signal frequency) under different frequency measurement noise levels.

Demonstrate the convergence behavior of your gradient-based algorithm in terms of the number of iterations required before convergence and the number of objective function evaluations needed in each line search.

Bonus: if you can implement another optimization algorithm, such as the particle swarm optimization (PSO), for performance comparison, you will receive 15% of your final report mark as bonus.

6.2.5 Guidelines for Demo

In your presentation, you need to cover at least item 1) from the progress report, as well as items 3), 4) and 5) from the final report.

In words, we expect to see your problem formulation, verification of gradient derivation, gradient-based algorithm design (i.e., the optimization strategy) and performance comparison with benchmark technique(s).

We encourage you to illustrate your design with various types of plots. As this project is in fact a 2-D optimization problem, visualization using contours (to show the modality of the objective function, satellite position, source position, etc), arrowed line segments to show the gradient direction and line search strategy, and RMSE curves a function of the FDOA measurement noise variance/standard deviation, can be used.

The marking scheme is similar to that of Project 1. We put emphasis on the presentation (20%) and graphic illustration (30%). The quality of your design (30%) and performance (20%) are also considered.

Chapter 7

Mathematics Background: Part II

7.1 Gaussian Distribution

Perhaps, the most widely used probability distribution is the multivariate Gaussian, also called multivariate normal (MVN) in some literature. It is mathematically convenient, and more importantly, Gaussianity is fairly reasonable in many applications.

Suppose $\mathbf{x} \in \mathcal{R}^{n \times 1}$ follows a multivariate Gaussian distribution, i.e., $\mathbf{x} \sim \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$, which is defined as

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{|2\pi\boldsymbol{\Sigma}|}} \exp\left(-\frac{(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}{2}\right). \quad (7.1)$$

Here, $\boldsymbol{\mu} = [\mu_1, \mu_2, \dots, \mu_n]^T = \mathbb{E}[\mathbf{x}]$ is the mean vector and $\boldsymbol{\Sigma} = \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T]$ is the covariance. The notation $|\cdot|$ represents the matrix determinant. In particular, the covariance matrix $\boldsymbol{\Sigma}$ is equal to

$$\boldsymbol{\Sigma} = \begin{bmatrix} \text{var}(x_1) & \text{cov}(x_1, x_2) & \cdots & \text{cov}(x_1, x_n) \\ \text{cov}(x_2, x_1) & \text{var}(x_2) & \cdots & \text{cov}(x_2, x_n) \\ \vdots & \vdots & \cdots & \vdots \\ \text{cov}(x_n, x_1) & \text{cov}(x_n, x_2) & \cdots & \text{var}(x_n) \end{bmatrix}, \quad (7.2)$$

where

$$\text{var}(x_i) = \mathbb{E}[(x_i - \mu_i)^2], \quad i = 1, 2, \dots, n, \quad (7.3)$$

and

$$\text{cov}(x_i, x_j) = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)], \quad i, j = 1, 2, \dots, n, \text{ and } i \neq j. \quad (7.4)$$

It can be shown (Exercise 3.4) that $\boldsymbol{\Sigma}$ is symmetric and at least positive semidefinite. When $n = 1$, (7.1) reduces to the univariate Gaussian distribution (Exercise 3.5),

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right). \quad (7.5)$$

One application of the Gaussian distribution is a transformation method popular in machine learning referred to as the re-parameterization trick. In some cases, the objective function is the expectation of another function $f(\mathbf{y})$ with respect to e.g., a Gaussian density $\mathcal{N}(\mathbf{y}; \boldsymbol{\mu}_y, \boldsymbol{\Sigma}_y)$, where the mean $\boldsymbol{\mu}_y$ and covariance $\boldsymbol{\Sigma}_y$ are the design variables to be optimized. The re-parameterization trick replaces $\mathbf{y}$ using

$$\mathbf{y} = \boldsymbol{\mu}_y + \mathbf{L}_y \boldsymbol{\epsilon}, \quad (7.6)$$

where $\Sigma_y = \mathbf{L}_y \mathbf{L}_y^T$ is the Cholesky decomposition of the covariance Σ_y , and ϵ follows a Gaussian density $\mathcal{N}(\epsilon; \mathbf{0}, \mathbf{I})$. As a result, the objective function becomes the expectation of $f(\mu_y + \mathbf{L}_y \epsilon)$ with respect to the Gaussian density $\mathcal{N}(\epsilon; \mathbf{0}, \mathbf{I})$ that is now independent of the design variables μ_y and Σ_y .

7.2 Maximum Likelihood Estimation

We are interested in estimating the parameter vector $\theta \in \mathcal{R}^{r \times 1}$ from measurements (also referred to as training data) in $\mathcal{D}$. The most commonly used approach for parameter estimation is to locate the parameter values that *assign* the highest probability to the measurements. This is called the maximum likelihood estimation (MLE), which indeed requires solving an optimization (maximization) problem. Mathematically, the MLE is defined as

$$\hat{\theta} = \max_{\theta} p(\mathcal{D}|\theta), \quad (7.7)$$

where $p(\mathcal{D}|\theta)$ is the distribution function of $\mathcal{D}$ parameterized by θ .

Sometimes, it is convenient to work with the logarithm of the objective function, which is $\log(p(\mathcal{D}|\theta))$. It can be shown (Exercise 3.6) that this transform does not change the solution to (7.7).

When the measurements in $\mathcal{D} = \{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_m\}$ are independent and identically distributed (i.i.d.), the logarithm of the original objective function becomes

$$\log(p(\mathcal{D}|\theta)) = \log \left(\sum_{i=1}^m p(\mathbf{y}_i|\theta) \right). \quad (7.8)$$

To illustrate the concept of MLE, consider the following example where the measurements are generated via

$$\mathbf{y} = [y_1, y_2, \dots, y_m]^T = \mathbf{G}\theta + \epsilon, \quad (7.9)$$

where $\mathbf{G}$ is a $m \times r$ measurement matrix. ϵ is the additive zero-mean Gaussian measurement noise (i.e., $\epsilon \sim \mathcal{N}(\epsilon; \mathbf{0}, \Sigma)$). It can be shown (Exercise 3.7) that the measurements in $\mathbf{y}$ follows a Gaussian distribution given by

$$p(\mathbf{y}|\theta) = \mathcal{N}(\mathbf{y}; \mathbf{G}\theta, \Sigma). \quad (7.10)$$

Note that in this case, the parameter vector θ determines the mean of the measurements, $\mathbf{G}\theta$, which according to (7.9), is also the unknown true values of the measurements.

To find the MLE of θ , we first notice that the logarithm of the objective function $p(\mathbf{y}|\theta)$ in (7.10) is, from (7.1),

$$\log(p(\mathbf{y}|\theta)) \propto -(\mathbf{y} - \mathbf{G}\theta)^T \Sigma^{-1} (\mathbf{y} - \mathbf{G}\theta), \quad (7.11)$$

where all the terms not depending on θ have been dropped. The MLE of θ for this particular example is thus given by

$$\hat{\theta} = \max_{\theta} \log(p(\mathbf{y}|\theta)) = \min_{\theta} (\mathbf{y} - \mathbf{G}\theta)^T \Sigma^{-1} (\mathbf{y} - \mathbf{G}\theta). \quad (7.12)$$

If the covariance of the noise ϵ , Σ , is a diagonal matrix with the diagonal elements being equal to one another (i.e., $\Sigma = \sigma^2 \mathbf{I}$), the MLE problem in (7.12) reduces to

$$\hat{\theta} = \max_{\theta} \log(p(\mathbf{y}|\theta)) = \min_{\theta} (\mathbf{y} - \mathbf{G}\theta)^T (\mathbf{y} - \mathbf{G}\theta). \quad (7.13)$$

This is in fact a linear least squares (LLS) problem. If Σ has non-identical diagonal elements and/or it is not a diagonal matrix, the MLE problem in (7.12) is a weighted least squares (WLS) problem in the sense that each measurement is not equally weighted in the objective function. Ideally, we can emphasize those measurements if we know they have better quality, to improve the estimation accuracy. It can be shown by using the results in Chapter 2.3 of the course notes (Exercise 3.8) that the MLE from (7.12) is

$$\hat{\boldsymbol{\theta}} = (\mathbf{G}^T \Sigma^{-1} \mathbf{G})^{-1} \mathbf{G}^T \Sigma^{-1} \mathbf{y}. \quad (7.14)$$

An important extension to the measurement model in (7.9) is that the true measurements are now *nonlinearly* related to the unknown parameters. Mathematically, we have

$$\mathbf{y} = \mathbf{g}(\boldsymbol{\theta}) + \boldsymbol{\epsilon}. \quad (7.15)$$

Here, the nonlinear measurement function $\mathbf{g}(\boldsymbol{\theta})$ is defined as

$$\mathbf{g}(\boldsymbol{\theta}) = [g_1(\boldsymbol{\theta}), g_2(\boldsymbol{\theta}), \dots, g_m(\boldsymbol{\theta})]^T, \quad (7.16)$$

where $g_i(\boldsymbol{\theta})$ produces the true value of the i -th measurement y_i , $i = 1, 2, \dots, m$.

In this case, the MLE of $\boldsymbol{\theta}$ is found by solving

$$\hat{\boldsymbol{\theta}} = \max_{\boldsymbol{\theta}} \log(p(\mathbf{y}|\boldsymbol{\theta})) = \min_{\boldsymbol{\theta}} (\mathbf{y} - \mathbf{g}(\boldsymbol{\theta}))^T \Sigma^{-1} (\mathbf{y} - \mathbf{g}(\boldsymbol{\theta})). \quad (7.17)$$

This is called the nonlinear least squares (NLS) problem. This formulation will be explored in the course project provide by Le Yang, it is an unconstrained optimization problem that can be solved by methods that was presented in Chapter 4 of this notes.

